

Alicia L. Bruzos, Dr.

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Marine Transmissible Cancers Research Group
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Trained in **biology** and **bioinformatics**, my education culminated in a **PhD in molecular medicine** focused on cancer genetics and evolution, later complemented by an additional master's degree in **high education teaching**. Over the last decade, I have built an **international research career** through positions in Spain, the United Kingdom, France, and Germany, and expanded my collaboration network through visiting stays in Portugal, South Korea, Ireland, and the United States.

As a molecular biologist, I am intrigued by the evolutionary convergence of cancers across different species, how malignant cells evolve to metastasize or 'travel to' distant tissues, organisms and, in some cases, even other species. My research focuses on **multi-omic approaches** in marine bivalves to uncover conserved mechanisms that underlie cancer transmissibility. Working with bivalves is not without challenges because they are not typically used in cancer genomics. The innovative approaches I have developed have generated breakthroughs that have advanced our understanding of transmissible cancers. These experiences converge in my current research line, which integrates **cancer genomics, marine ecology and microbiology**.

PERSONAL DETAILS

Date of birth: 29-March-1993 (33 years old)

Nationality: Spain

EDUCATION

08/09/2022 **PhD in Molecular Medicine**, U. Santiago de Compostela, SPAIN. International & Cum laude¹.

18/02/2025 **MSc in University Teaching**, Universidad Europea de Madrid (*online*), SPAIN.

29/07/2016 **MSc in Bioinformatics**, Universitat Autònoma de Barcelona, SPAIN.

31/07/2015 **BSc in Biology**, Universidade de Santiago de Compostela, SPAIN. 1-year Erasmus fellow at Université Libre de Bruxelles, BELGIUM.

Certified language levels*: **English C1; French B2; Galician C1; Spanish C2 (native); German A1.**

ACADEMIC POSITIONS

Sep 2025 – ongoing **Junior Group Leader** (*Minerva Fast Track fellowship*). Max Planck Institute for Marine Microbiology, GERMANY. Symbiosis Department. My team is made up of 1 Postdoc, 1 PhD student, 1 Technician and eventually MSc/BSc students.

Jul 2023 – Aug 2025 **Postdoctoral researcher** (*Marie Skłodowska-Curie COFUND fellowship*). Université de Caen Normandie, FRANCE. PI: Camille Détrée and Pascal Claquin. BOREA lab.

Sep 2021 – Apr 2023 **Postdoctoral researcher**². The Francis Crick Institute, University College London, UNITED KINGDOM. PI: Veronica Kinsler. Mosaicism and Precision Medicine lab.

Sep 2016 – Sep 2021 **Predocctoral student** (*Spanish national predocctoral fellowship*). U. de Santiago de Compostela, SPAIN. PhD advisor: Seila Díaz. Mobile Genomes lab led by Jose Tubio.

SHORT-TERM RESEARCH STAYS AS A VISITING SCIENTIST (totalling 11 months)

- **Centro de Ciências do Mar (CCMAR)**, Algarve, PORTUGAL. Jul-Aug 2024.
- **Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, KOREA REP. Jan-Apr 2021.
- **Universidade de Vigo**, Vigo, SPAIN. Sep-Dec 2020.
- **University of Galway**, Galway, IRELAND. Mar-Apr 2019.
- **Pacific Northwest Research Institute (PNRI)**, Seattle, USA. Sep-Dec 2018.
- **Université Libre de Bruxelles**, Brussels, BELGIUM. Jun-Jul 2014.

¹In Spain, doctoral theses graded as excellent may also be awarded a '*cum laude*' mark which requires unanimous agreement among the members of the evaluation board (secret ballot). ² Starting date (Sep 2021), a year earlier of doctoral defense (Sep 2022) due to fellowship ending, postdoc contract with a salary reduction until being a doctor.

*According to *The Common European Framework of Reference for Languages*.

SCIENTIFIC PRODUCTION

ORCID: [0000-0003-4362-545X](https://orcid.org/0000-0003-4362-545X) ResearcherID: [D-7235-2018](https://www.researcherid.com/rid/D-7235-2018) ~ [GScholar](https://www.gscholar.org/), [WOS](https://www.wos.org/), [ResearchGate](https://www.researchgate.net/), [Scopus](https://www.scopus.com/).

I began my research career in **human cancer genomics** within the *International Cancer Genome Consortium*, where I analysed retrotransposon insertions and viral integrations that destabilise chromosomes and contribute to tumorigenesis [7, 8, 9]. This foundation shaped my perspective on genome instability and cancer evolution. During my doctoral work, I turned to **transmissible cancers in marine bivalves**, integrating large-scale field sampling with histopathology, molecular assays, and genomics. I co-discovered horizontal mitochondrial transfer, an early whole genome duplication, co-infection by multiple cancer lineages, amplification of well-known oncogenes revealing the evolutionary parallels between marine cancers and metastasis [4]. Furthermore, I co-discovered a cross-species transmission of cancer between two clam species, broadening our understanding of metastatic barriers [5]. In my first postdoctoral work in the UK, I returned to **human precision medicine**, co-discovering mosaic *BRAF* and *RAF1* fusions as recurrent drivers of congenital naevi that sometimes progress to cancer [3]; this work laid the foundation to MEK inhibitors treatments that have been tested on patients showing promising results. For my second postdoctoral work in France, I then designed a project on the **environmental dynamics of transmissible cancers**, designing a PCR-based assay for routine monitoring [2] and characterising their spatio-temporal spread in cockles and co-habiting species [1]. These experiences now converge in my independent research group in Germany, which integrates cancer genomics and marine biology to explore **microbiome-cancer interactions** and to identify **conserved mechanisms of metastasis across species**.

To date, I have authored **13 peer-reviewed international Q1 articles** (11 of which are D1), and **10 book chapters** with ISBN (6 of which as first author). Notably, I am **first author on 5 articles**, the **corresponding author on 2**, and **7 are not co-authored by my thesis advisor**, reflecting my capacity for autonomous research. My work has garnered over **2,200 accumulated citations**.

SELECTED SCIENTIFIC PUBLICATIONS²

1. Bruzos, AL^{1/7**} et al. (2025) *Prevalence of hemic neoplasia in cockles and co-habiting bivalve species in the western coast of France and the southern coast of Portugal*. bioRxiv: doi.org/10.1101/2025.07.30.667629. Accepted for publication on **Philosophical Transactions of the Royal Society B**.
2. Santamarina, M*; Bruzos, AL^{2/5*†}; Pequeño, A; Rodríguez, J; Díaz, S. (2024) Novel PCR molecular assay for the identification of marine transmissible cancers from *C. edule*. In: **Journal of Invertebrate Pathology**, 207: 108232. doi.org/10.1016/j.jip.2024.108232 (OA).
3. Martín, SB*; Polubothu, S*; Bruzos AL^{3/18*}; ... Kinsler, VK. (2023) *Mosaic BRAF fusions are a recurrent cause of congenital melanocytic naevi targetable by MEK inhibition*. In: **Journal of Investigative Dermatology**, 144(3):593-600. doi.org/10.1016/j.jid.2023.06.213 (OA).
4. Bruzos, AL^{1/45*}; Santamarina, M*; ... Tubío, JMC. (2023) *Somatic evolution of marine transmissible leukemias in the common cockle, *Cerastoderma edule**. In: **Nature Cancer**, 4:1575-1591. doi.org/10.1038/s43018-023-00641-9 (download here). Issue cover.
5. Garcia-Souto, D*; Bruzos, AL^{2/20*}; ... Tubío, JMC. (2022) *Mitochondrial genome sequencing of marine leukemias reveals cancer contagion between clam species in the Seas of Southern Europe*. In: **eLife**, 11:e66946. doi.org/10.7554/eLife.66946 (OA).
6. Ricardo, F et al. (includes Bruzos, AL^{3/8}). (2021) *Assessing the elemental fingerprints of cockle shells to confirm their geographic origin from regional to international spatial scales*. In: **Science of the Total Environment**, 814:152304. doi.org/10.1016/j.scitotenv.2021.152304 (download here).
7. Álvarez, EG et al. (includes Bruzos, AL^{13/43}). (2021) *Aberrant integration of Hepatitis B virus DNA promotes major restructuring of human hepatocellular carcinoma genome architecture*. In: **Nature Communications**, 12:6910. doi.org/10.1038/s41467-021-26805-8 (OA).
8. Rodríguez-Martin, B et al. (includes Bruzos, AL^{15/53}). (2020) *Pan-Cancer analysis of whole genomes identifies driver rearrangements promoted by LINE-1 retrotransposition*. In: **Nature Genetics**, 52:306-319. doi.org/10.1038/s41588-019-0562-0 (OA).
9. The ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium (includes Bruzos AL, position 60 of 1340 authors). (2020) *Pan-Cancer analysis of whole genomes*. In: **Nature**, 578:82-93. doi.org/10.1038/s41586-020-1969-6 (OA). Issue cover.



² Only a selection is listed, [check here the full list](#). The * denotes first authorship (2, 3, 4 and 5); ⁰⁰ denotes (my position / total number of authors); † =corresponding author (2); OA=open access.

CONFERENCES

I have presented my research at **53 national and international conferences** across **twelve countries**. This includes **6 invited talks**, **22 oral presentations** delivered by myself, **25 poster presentations** (15 of which I personally presented), and **6 award-winning contributions** (3 for best talk and 3 for best poster). Additionally, I have served on the scientific and organizing committees of **3 conferences** in Spain, the UK, and France. Beyond conferences, I have been invited to deliver **9 core seminars** at scientific institutions in 4 different countries. Detailed list [here](#).

AWARDS AND HONORS

I have been recognised with several distinctions for both research excellence and outreach. I was awarded the prestigious **L'Oréal-UNESCO 'For Women in Science' Young Talents Award** (20,000€, France, 2024) and was a finalist for the **W.D. Hamilton Award** of the Society for the Study of Evolution (500\$, USA, 2023). My work in public engagement has also been acknowledged, including the **Outreach Award in Medicine and Health** (Fundación Lilly & The Conversation, 2023), the **3MT Video Award** (Cuéntame11F Spain, 2022), and the **Women's Name Award for Scientific Vocations** (Fundación Merck Salud Spain, 2022). In addition, I received 6 prizes for research presentations at national and international meetings. Overall, I have been a finalist or awardee in 14 competitions since 2013, spanning scientific innovation, communication, and creative initiatives.



FUNDING

I have consistently secured competitive fellowships and project funding throughout my career, including at the doctoral, postdoctoral, and independent group-leader stages. Beyond these, I obtained travel grants and contributed to international projects, reflecting my ability to build collaborations and pursue ambitious research. This sustained record of acquiring my own funding demonstrates my progression towards scientific independence and leadership.

A) Competitive fellowships:

- 2024 **Minerva Fast Track Fellowship (2025-2028)**. Max Planck Society, GERMANY. *ONGOING*.
- 2023 **National Postdoctoral 'Juan de la Cierva' Fellowship (2024-2026)**. Spanish Ministry of Science (ref. JDC2022-048396-I), SPAIN. *DECLINED to accept the MSCA in France*.
- 2023 **EU MSCA Fellowship (2023-2025)**. Marie Skłodowska-Curie Actions (MSCA-cofund) and Région Normandie (ref. 101034329), EU and FRANCE. *CONCLUDED*.
- 2017 **National Doctoral Fellowship (2017-2021)**. Spanish Ministry of Economy, Industry and Competitiveness (ref. BES-2016-078166), SPAIN. *CONCLUDED*.

B) Research projects: I have successfully secured over 40,000€ of funding from **five international competitive grants**, which evidences my progression towards scientific autonomy. This included an *Exploration Fund Grant* (5,000 €) from The Explorers Club (USA) in 2024; a *FOVI project* (ref. FOVI230047; 30,000 €) in 2023 as a collaborator from Gobierno de Chile; two *EU transnational access* TNA grants (together over 10,000 €), one for Ireland in 2019 (Assemble+) and the other one for Portugal in 2024 (AquaExcel3.0); and a BIF grant (2,370 €) from Boehringer Ingelheim Fonds (Germany) in 2021.

C) Conference grants: I have received **seven grants from major international societies** (including SMBE, EACR, FEBS, MCAA, among others), enabling my participation in global meetings across Europe and Asia.

MENTORING OF STUDENTS

I have supervised **seven undergraduate and master's students** during their theses in Germany, France and Spain, three of them subsequently pursued a PhD after, and one received an award for the quality of their thesis. I have also supervised **three internships**, two of which obtaining Erasmus+ funding.

I am currently **supervising three students**: the PhD student Nupur Potukuchi from University of Bremen (Germany), the MSc student Nur M Farhad from University of Basque Country (Spain) and the Bachelor intern Maxim Hess from University of Bremen (Germany). All of them as the sole supervisor without the involvement of a senior co-supervisor. More details [here](#).

HIGH EDUCATION TEACHING

I have gained teaching experience (**134.5 certified teaching hours**) at **six public universities across four countries** (Spain, UK, France, and Chile), delivering theoretical and laboratory sessions in **different languages** (English, Spanish/Galician, and French) to **undergraduate and master's students**. Highlights include lecturing in the MSc *Sciences de la Mer* program at U. de Caen Normandie (2023–2025), teaching medical students at

U. College London (2021–2023), contributing to genetics courses at U. de Santiago de Compostela (2017–2021), and recent contributions at the U. de O'Higgins (7.5 hours, 2024 with the FOVI project). **These experiences have improved my ability to adapt to diverse academic settings** and I also hold national accreditations as an assistant professor in Spain (PAD ANECA) and France (Section 64).

OUTREACH ACTIVITY

I have actively participated in over 40 public science outreach events, including 16 school visits (Spain, UK, France), 12 popular science articles (*e.g.*, The Conversation), 8 talks for the general public (*e.g.* *Pint of Science* in London, A Coruña, and Caen), 6 promotional videos (*e.g.*, AquaExcel project), 5 workshops on scientific fairs, 3 opinion editorials, and 5 miscellaneous activities such as contributing to children's science colouring book or the school radio program of IES Mata Jove. **I am strongly committed to making science accessible, particularly in towns where opportunities to engage with scientists are limited.**

SERVICE TO THE COMMUNITY

- **Conference organization.** I have been involved in **organizing 3 conferences**: the VIII Young Researchers Meeting (2021), with 300 attendees, 135 oral presentations, and 67 posters (we edited an abstracts book with ISBN) hosted by USC (Spain); the 5th Crick Postdoc Symposium (2022), which brought together 160 participants for 2 plenary talks, 11 short oral presentations, and 38 posters (we gave 4 prizes) hosted by The Crick (UK); and the Chilean Coastal Ecological Health Symposium (2024), with 30 attendees, 4 plenary talks, and a panel discussion; hosted by UNICAEN (France).
- **Journal reviewing activity.** I have been invited to review nine manuscripts for the international community in highly-recognized journals: 1 *Science*, 1 *Molecular Ecology*, 2 *Science Advances*, 3 *Proc. Biol. Sci.*, 1 *Scientific Rep.*, 1 *Journal of Evolutionary Biology* (see verified reviews on Publons: D-7235-2018).
- **Member of juries and evaluation committees.** External evaluator for two PhD theses (L. Tomás, University of Vigo, 2024; A. Guitián Caamaño, University of A Coruña, 2024); substitute member of a PhD examination committee (S. Blanco González, University of Vigo, 2025). Panel member for the MSCA-COFUND Iberus InterDoc Programme (Health Technologies and Agrofood & Nutrition panels, 2nd Call 2026), evaluating 18 doctoral applications. In 2026, I was invited to review in round 1 the BBSRC standard research grant application for the UK Research and Innovation Public Funding Service. Jury member of the Galicia Spin-Off Awards (Gciencia, 2024).
- **Institutional responsibilities.** I have served as an elected spokeswoman for doctoral students on the committees of the International Doctoral Schools at *Universidade de Santiago de Compostela* (2021-2022) and *Universidade de Vigo* (2017-2018). In Vigo, I was also part of its Quality Commission.
- **Professional membership:** I am an active member of the *Marie Curie Alumni Association* (MCAA) since 2023 and the *European Association for Cancer Research* (EACR) since 2019.

RELEVANT CERTIFICATIONS

I hold a **first-aid certification** (2023) from the Université de Caen Normandie, focused on fieldwork-related emergencies. I am certified in **biological risk management** (UNE-CWA 15973, 2017) and in **animal experimentation procedures** (A+B+C+D+E, 140 hours, 2017), including practical training with fish and rodents, accredited by the Comunidad de Madrid. My field training includes a **Craft Skipper License** (PER, 60 hours, 2017), combining theoretical and practical modules (engine, VHF, sailing); an **Open Water Diver License** (60 hours, 2014); and a certification of **Outdoor Activity Instructor** by Consellería de Traballo da Xunta de Galicia (350 hours, 2012), which involved extensive training in group coordination and safety in outdoor settings.

MEDIA COVERAGE

PR clipping: albruzos.github.io/media

Renowned media have featured my research publications and various scientific activities (outreach events, awards, or career steps) in over 250 news items published in more than 40 media platforms across 10 countries. This includes *BBC radio*, *The New York Times*, *El País* or *La Voz de Galicia*, as well as in TV newscasts from 2 countries (*France3normandie*, *Televisión de Galicia*, *T. Princ. Asturias*, *Canal Sur*). Beyond promoting my research, I am often invited to share my opinion on scientific careers, current topics, or to advocate for women in science. **These initiatives highlight my dedication to science communication and my leadership in public engagement.**

SOCIAL MEDIA:  [@BruzosAliciaL](#)  [@BruzosAliciaL](#)  [@alicialbruzos](#)  [@BruzosAliciaL](#)

This trajectory positions me uniquely to lead the research that brings together immunity, microbiota, and environment to establish an ecological and evolutionary framework for transmissible cancers.